Gravity from Quantum Measurement

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Abstract

We propose a framework in which gravity can emerge from the physical backreaction of quantum measurement and the microstate dynamics of information acquisition. Standard results, including horizon scales and entropy bounds, follow from Landauer's principle at the Unruh temperature. We then present a concrete bilocal microstate construction as a case-study. Entangled pairs emitted from a localized source propagate into left and right Rindler wedges, where polarization measurement events produce discrete energy-momentum impulses and correlated classical bits. These microstates can be described geometrically using Aichelburg–Sexl pp-wave solutions, whose longitudinal shifts and transverse focusing imprint energy and information directly onto horizon geometry. This setup illustrates how individual measurement events contribute to horizon entropy, and how entanglement can be converted into spatial information, providing an operational realization of ER=EPR at the level of single microstates. The framework complements entropic and thermodynamic approaches by supplying a concrete, microphysical model linking quantum measurement, horizon dynamics, and emergent spacetime.

1 Introduction

Two landmark results establish that gravity and thermodynamics are deeply intertwined. Jacobson showed that the Einstein equations can be derived from thermodynamic identities applied to local Rindler horizons, with horizon entropy playing the role of the fundamental quantity from which spacetime dynamics emerge [1]. Verlinde extended this perspective, arguing that gravitational forces arise from entropy gradients on holographic screens, recasting gravity as an entropic rather than a fundamental interaction [2]. Both frameworks, however, operate at the level of bulk thermodynamics and leave open a prior question: what physical process underlies each individual entropy increment?

We argue that the answer is quantum measurement. Each act of information acquisition by a physical observer constitutes a discrete microstate event, and the collective backreaction of such events—together with the causal correlations they induce—gives rise to gravitational effects. The argument proceeds in three steps. First, any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. Second, by the equivalence principle, this impulse is locally indistinguishable from the effect of a gravitational field: no local measurement can distinguish the recoil associated with a quantum measurement event from free fall in a gravitational potential. Third, when such impulses are modelled geometrically—via Aichelburg–Sexl (AS) pp-wave solutions, as we show—they imprint energy and information directly onto horizon geometry,

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and coarse-graining over the resulting ensemble recovers standard horizon thermodynamics.

A unifying structure runs through all three of these relations, and it is worth making explicit before proceeding. In each case, the relation between two adjacent concepts has the same logical character: every instance of the more abstract concept is grounded in an instance of the more physical one, but not every instance of the more physical concept gives rise to the more abstract one *for a given observer*. The gap in each case is filled by subjectivity. Quantum state reduction is always backed by an observation—it just is a Bayesian update on the observer's part—but an observation that carries no new information for that observer produces no state reduction. Equally, every observation is backed by an energy-momentum transfer, but not every such transfer constitutes an observation for you: an interaction involving a different observer leaves your state unchanged. The Unruh effect suggests the same structure extends to the equivalence principle: an accelerating observer perceives a thermal bath and attributes it to a gravitational field, while an inertial observer sees only vacuum. Whether an acceleration counts as gravity is perspectival in precisely the same sense. In each case the relation is an identification, not a causal link, and the identification is observer-relative.

This stands in sharp contrast to the Penrose–Diósi objective collapse programme [3, 4], which postulates that gravity *causes* quantum state reduction as an objective, observer-independent physical process. Figure 1 summarises this contrast. The three solid arrows represent the perspectival grounding relations described above; the single dashed arrow represents the Penrose–Diósi causal postulate. It is the only relation in the diagram claimed to be objective and observer-independent, and it is the only one we do not adopt.

$$\begin{tikzcd}
\text{Gravitational field} \arrow[r, dashed, "\text{Penrose–Diósi}"] & \text{Quantum state reduction} \\
\text{Acceleration} \arrow[u, "\text{equivalence principle}"] \arrow[r, "\text{energy–momentum transfer}"'] & \text{Observation} \arrow[u, "\text{Bayesian update}"']
\end{tikzcd}$$

Figure 1: Solid arrows denote substrate relations: $A \to B$ means every instance of B is physically realised by an instance of A, for a given observer. The dashed arrow denotes the Penrose–Diósi postulate, where the relation is claimed to hold objectively rather than observer-relatively.

To make this concrete, we build on the microstate construction of [5], in which bilocal driving sources generate entangled Rindler excitations that realise individual terms of the Unruh thermal ensemble. The present paper applies and extends that construction. Entangled pairs emitted from a localised source propagate into left and right Rindler wedges; polarisation measurement events produce discrete energy-momentum impulses and correlated classical bits. We show that each such microstate carries a calculable gravitational imprint, modelled by an AS pp-wave solution, and that coarse-graining over the ensemble recovers standard horizon thermodynamics. This construction also provides an operational realisation of ER=EPR at the level of individual microstates: entanglement is converted into geometric information, and each measurement event contributes a discrete, calculable increment to horizon entropy.

This approach both complements and sharpens the thermodynamic gravity programme of Jacobson and Verlinde. Jacobson's derivation of the Einstein equations from horizon thermodynamics and Verlinde's entropic force framework both operate at the level of

bulk averages, leaving open the question of what physical process underlies each entropy increment. The present construction supplies that microstate realisation: each entropy increment corresponds to a physically mediated measurement event with a calculable energy-momentum transfer. Thermality is emergent rather than intrinsic, arising from coarse-graining over an ensemble of source-driven microstates, each of which admits a definite, non-thermal description in the global Minkowski frame. The thermodynamic results of both programmes are recovered as a consequence of this coarse-graining, rather than assumed as a starting point.

[Section map paragraph todo]

2 Motivating Heuristics

The central observation is simple. Landauer's principle associates a minimal energy cost with the acquisition of information [6]. For an accelerated observer, the relevant temperature is the Unruh temperature [7]. Substituting one into the other eliminates temperature entirely, yielding a frame-independent relation between information acquisition, energy transfer, and acceleration (a version of the BHT first law). This is not merely an algebraic convenience: it is a concrete dictionary between the thermal description natural to a Rindler observer and the non-thermal global Minkowski description, with the apparent thermality of the vacuum emerging as a consequence of the observer's acceleration rather than as a fundamental property of the state. In this way we derive a sequence of standard results in black hole thermodynamics.

2.1 The Unruh-Landauer Relation

Landauer's principle states that the acquisition of ΔS bits of information by a physical observer requires a minimum energy transfer

$$\Delta E \geq k_B T \, \Delta S. \tag{1}$$

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta S.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a frame-independent relation between energy, acceleration, and information, with no reference to any thermal description. It can be read as the first law of black hole thermodynamics derived from below — from the physical cost of information acquisition — rather than postulated as a thermodynamic identity. In what follows, all standard results emerge as consequences of this single relation. We note that Landauer's principle is conventionally stated for information erasure rather than acquisition. Here we apply it to acquisition, which is the physically relevant process for an observer registering a measurement outcome. The bound has the same form in both cases, and we proceed with the understanding that this is a heuristic application whose precise justification is deferred to the microstate construction of Section 3.

2.2 Momentum Transfer and Acceleration

The energy transfer ΔE implied by equation (3) is carried by a physical quantum — matter or radiation — mediating the measurement interaction. We refer to this as the

thrust quantum. For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{4}$$

Let the interaction occur over a spatial scale r, so that the interaction time is $\Delta t \sim 2r/c$. For a detector of mass m_d, Newton's second law in impulse form gives

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2 m_d r}. \tag{5}$$

The factor of 2 in the interaction time is not an independent assumption: it will be fixed self-consistently by requiring that the characteristic interaction scale reproduces the Schwarzschild radius, as shown in the next subsection.

2.3 The Schwarzschild Radius

The thrust quantum carries energy ΔE and produces an acceleration a on the detector. Identifying this with the Newtonian gravitational acceleration due to a mass m_t at distance r.

$$a = \frac{G m_t}{r^2}, \tag{6}$$

and substituting equation (5) with $\Delta E = m_t c^2$, we obtain

$$\frac{m_t c^2}{2 m_d r} = \frac{G m_t}{r^2}. \tag{7}$$

Solving for r:

$$r = \frac{2 G m_d}{c^2} \equiv r_s, \tag{8}$$

which is precisely the Schwarzschild radius of the detector mass m_d. This fixes the factor of 2 in the interaction time retroactively: the choice $\Delta r = 2r/c$ is the unique one consistent with Newtonian gravity and the correct Schwarzschild scale.

2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{G m_d}{r_s^2} = \frac{c^4}{4 G m_d}, \tag{9}$$

in agreement with the standard Schwarzschild result. This is the acceleration that appears in the Unruh temperature associated with the detector's horizon, $T = \hbar\kappa / 2\pi k_B c$, providing an internal consistency check on the framework.

2.5 Bekenstein-Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity κ for a, the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta S \leq \frac{2\pi c}{\hbar \kappa} \Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot c^2 \Delta m_d. \tag{10}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta S = \frac{\Delta A}{4 \ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{11}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

2.6 Summary

Beginning from the single relation (3), and using only Newton's law and the equivalence principle, we have recovered the Schwarzschild radius, the surface gravity, and the Bekenstein-Hawking area law as consequences of the observation=acceleration identification. These are standard results, but their derivation here from the physical cost of information acquisition rather than from thermodynamic postulates is the point. The thrust quantum introduced in this section — the physical carrier of the measurement-induced impulse — will be given a precise microstate realization in the sections that follow, where it is identified with an individual photon modeled as an Aichelburg-Sexl pp-wave.

3 Rindler/Unruh Microstates from a Bilocal Source

Building on the local Rindler-frame heuristics of measurement-induced acceleration, we provide an explicit microstate construction that realizes the Unruh effect via source-driven excitations. In standard treatments, a uniformly accelerated observer perceives the Minkowski vacuum as a thermal bath at temperature

$$T = \frac{a}{2\pi}, \tag{12}$$

where a is the proper acceleration; this thermality arises from tracing out one Rindler wedge of an entangled vacuum. Our approach complements this by showing that the apparent thermal response can be realized through physically localized, bilocal sources that generate entangled particle pairs in left and right Rindler wedges.

A bilocal driving source couples to Unruh modes, producing states

$$|\Psi_\omega\rangle = |1_\omega\rangle_L \, |1_\omega\rangle_R, \tag{13}$$

which correspond to individual terms in the thermal ensemble. Measurement by an accelerated observer projects onto these microstates, producing a local energy-momentum impulse and registering a horizon entropy increment. In this sense, the "thermal" response is not solely a property of the vacuum, but emerges from the observer's interaction with physically mediated microstates.

We further refine the construction to control localization: by restricting Rindler modes to smaller wedges, intersecting wedges, or Gaussian-enveloped wave packets, the excitations can be made compact in space and time, smoothly interpolating between global thermality and sharply peaked, non-thermal microstates. This demonstrates that portions of the Unruh effect can be understood operationally, with acceleration-induced detection corresponding to discrete, physically realizable microstate events.

This framework connects directly to emergent gravity: each detection event contributes a discrete energy-momentum impulse, producing a local acceleration which, via the equivalence principle, manifests as a gravitational effect. By summing over many such events across horizons, the familiar global gravitational and thermodynamic structures—Schwarzschild radius, Bekenstein bound, and horizon entropy—emerge naturally from the microphysical dynamics of information acquisition.

4 pp-Wave Microstates and Spacetime Response

To model the gravitational effect of a single photon, we consider the Aichelburg–Sexl pp-wave solution in general relativity. Let

$$u = t - z, \quad v = t + z, \tag{14}$$

where the photon propagates along the null plane $u = 0$, z is the longitudinal direction, and x, y are transverse coordinates.

Consider two test particles initially at rest at $y = 0$, $z = 0$, with fixed x coordinates, before the photon arrives ($u < 0$). As the photon crosses $u = 0$ (the shock wave), the spacetime geometry produces two distinct effects on the test particles:

1. **Longitudinal shift:** The v coordinate is displaced by an amount proportional to $\ln(r)$, where $r^2 = x^2 + y^2$. Near the center ($x \to 0$), the shift becomes very large, effectively dragging the particle along at nearly the speed of light. During this process, no proper time elapses for the particle crossing the shock.
2. **Transverse kick:** The gradient of the shift produces a transverse velocity toward the center of the wave, proportional to $1/r$ (or $1/x$ for $y = 0$). This represents a focusing effect, pulling the particles toward the axis of propagation.

From the perspective of emergent spacetime, this microstate illustrates how localized energy-momentum (here, a single photon) induces both longitudinal and transverse responses in the geometry. The shift in v encodes a lightlike dragging effect, while the transverse kick produces spatial correlations between initially separated test particles. In our framework, such pp-wave interactions provide a **microstate-level mechanism** by which entangled or energetic quanta imprint spacetime structure, complementing the Rindler/Unruh construction and connecting directly to horizon microphysics.

5 Microstate Geometry and pp-Wave Realization of Horizon Entropy

To make the preceding heuristic arguments concrete, we construct an explicit microstate realization using a bilocal source setup reminiscent of the Unruh-driving framework [5]. Consider a past emitter, *Paz*, who releases an entangled photon pair in the Bell state

$$\frac{1}{\sqrt{2}}\big(|H\rangle \otimes |V\rangle + |V\rangle \otimes |H\rangle\big), \tag{15}$$

with equal and opposite momenta $\pm\mathbf{k}$. Initially, the photons are localized at Paz and carry no net momentum, so there is no backreaction on the emitter. As the photons propagate, they cross the past Rindler horizons into the left and right Rindler wedges, where two observers, *Lester* and *Wright*, record their polarization outcomes in the H/V basis. Each measurement produces a classical bit, which is perfectly correlated between the observers.

Crucially, a future observer may instead ask the question: "Who observed H?" The answer is a spatial bit, distinguishing left vs. right wedge. In this sense, the polarization information carried by the photons is converted into *spatial information*, providing a direct microphysical mechanism for emergent space. This realizes ER=EPR at the level of single Unruh microstates, with the Rindler horizon acting as the substrate where information is physically encoded.

To capture the gravitational imprint of each microstate, we model each photon as an Aichelburg–Sexl pp-wave propagating along the null plane $u = t - z = 0$ in Minkowski coordinates. Test particles at fixed transverse positions (x, y) experience the pp-wave as a shock:

- A longitudinal shift $\Delta v \sim \ln r^2$, with $r^2 = x^2 + y^2$, which drags particles along the lightlike direction without generating proper time.
- Transverse kicks proportional to $\nabla \ln r$, focusing particles toward the wave center.

These effects encode the energy-momentum of the photon into the horizon geometry, identifying the transverse plane of the pp-wave with horizon area elements $dA = dx\,dy$. Each detection thus imprints a calculable contribution to the horizon entropy, with

$$\Delta S \sim \frac{8\pi G m_{\text{photon}}}{\hbar c^3}\,\Delta E, \tag{16}$$

matching the energy-entropy relation derived heuristically in Section **??**.

In a steady-state emission scenario, continuous streams of entangled photons maintain the accelerations of Lester and Wright, and coarse-graining over many microstates recovers the familiar thermal Unruh spectrum. Thermality is therefore *emergent*, not intrinsic: it arises from averaging over a large ensemble of source-driven microstates, each of which has a well-defined energy-momentum and horizon imprint.

This framework provides a concrete link between microstate-level dynamics, observer-dependent accelerations, and horizon thermodynamics. Each measurement-induced impulse has both a local physical effect (acceleration of the detector) and a global geometric imprint (contribution to horizon area and entropy), formalizing how the backreaction of quantum measurement can generate gravitational phenomena.

6 Emergent Proper Time

How can we quantize gravity without quantizing spacetime (or at least curvature)? And how can we do QFT computations in a probabilistic discrete spacetime? It may be that these issues must be dealt with when trying to understand quantum gravity. One may take the viewpoint that spacetime is made up of a mesh or sprinkling of Planck scale lengths and times, but this is not immediately Lorentz invariant and seems to side step the issues [8]. We instead take the viewpoint that space and time are actually properties of matter; that space and time only makes sense up to the resolution of the matter emanating it. Using action to do this grants us automatic Lorentz invariance. The approach also allows us to define a system where emergent proper time and space can be compared to gravity at classical scales. We show how gravity is interpreted as the mechanics of entangled spacetimes.

Proper time along a particle's worldline can be interpreted as a sequence of discrete "ticks" of action, each of magnitude $\hbar$, representing a minimal quantum of evolution. This construction is Lorentz-invariant, naturally connects to the phase factor $e^{iS/\hbar}$ in the path integral formulation, and can be understood as matter being sampled by the Higgs field or other persistent interactions. In this sense, massive particles act as a form of physical memory, while massless particles primarily carry information without enduring action quanta.

Classical spacetime and particle trajectories emerge from the constructive interference of these ticks across all possible paths, reproducing smooth motion in the classical limit. The maximal rate at which these ticks can occur is bounded by the particle's mass-energy, consistent with the Bremermann or Nyquist limit, imposing a fundamental constraint on the flow of proper time. Viewed in this way, spacetime is not a fundamental entity, but rather an emergent structure arising from the informational content encoded in the discrete action of matter.

7 Emergenct Space: ER=EPR

Consider a bilocal source, *Paz*, located in the past. She emits an entangled photon pair in the polarization Bell state

$$\frac{1}{\sqrt{2}}\big(|H\rangle \otimes |V\rangle + |V\rangle \otimes |H\rangle\big), \tag{17}$$

where H and V denote horizontal and vertical polarizations. Initially, the two photons are co-located at Paz, and the pair carries no net momentum, so there is no backreaction on the source.

The photons propagate across the past Rindler horizons into the left and right Rindler wedges, becoming microstates responsible for the Unruh-type acceleration experienced by the left observer, *Lester*, and the right observer, *Wright*. Each observer detects the photons in the H/V polarization basis, recording classical bits of information in their respective memories. These recorded bits are perfectly correlated due to the initial entanglement.

Now, consider a future observer asking not "what polarization was measured?" but rather "which observer recorded the H polarization?" The answer is a spatial label, left or right, effectively converting polarization information into spatial information. In this way, the classical correlations induced by the entangled photons produce emergent spatial structure, linking previously co-located degrees of freedom to distinct positions in the left and right wedges. This mechanism realizes an explicit microstate version of the ER = EPR principle within the Unruh framework: space itself emerges from the entanglement and measurement of localized quantum microstates.

8 Conclusion

9 Acknowledgements

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